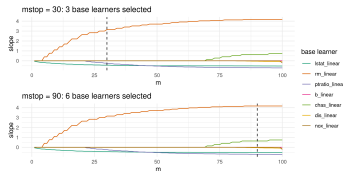
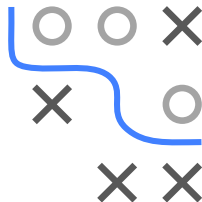


Introduction to Machine Learning

Gradient Boosting: CWB Basics 2



Learning goals

- Handling of categorical features
- Intercept handling
- Practical example

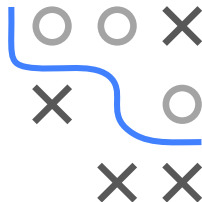
HANDLING OF CATEGORICAL FEATURES

Advantages of simultaneously handling all categories in CWB:

- Much faster estimation compared to using individual binary BLs
- Explicit solution of $\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^G} \sum_{i=1}^n (y^{(i)} - b_j(x_j^{(i)} | \theta))^2$:

$$\hat{\theta}_g = n_g^{-1} \sum_{i=1}^n y^{(i)} \mathbb{1}_{\{x_j^{(i)} = g\}}$$

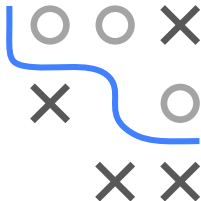
- For features with many categories we usually add a ridge penalty



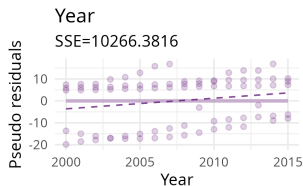
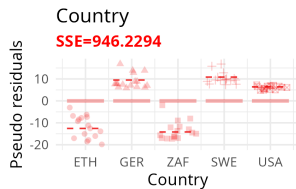
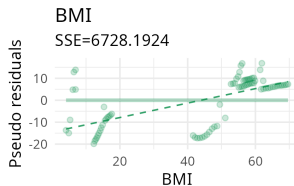
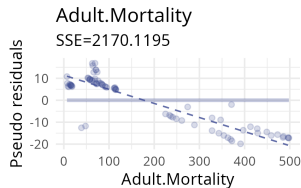
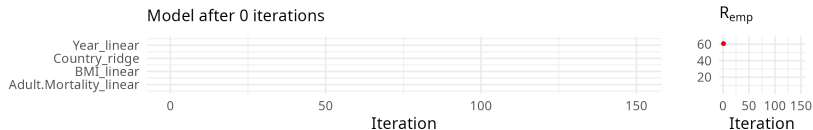
INTERCEPT HANDLING

There are two options to handle the intercept in CWB. In both, the loss-optimal constant $f^{[0]}(\mathbf{x})$ is an initial model intercept.

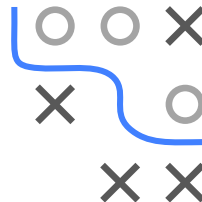
- 1 Include an intercept BL:
 - Add BL $b_{\text{int}} = \theta$ as potential candidate considered in each iteration and remove intercept from all linear BLs, i.e., $b_j(\mathbf{x}) = \theta_j x_j$.
 - Final intercept is given as $f^{[0]}(\mathbf{x}) + \hat{\theta}$. Linear BLs without intercept only make sense if covariates are centered (see [mboost tutorial](#), p. 7)
- 2 Include intercept in each linear BL and aggregate into global intercept post-hoc:
 - Assume linear base learners $b_j(\mathbf{x}) = \theta_{j1} + \theta_{j2} x_j$. If base learner $\hat{b}_j$ with parameter $\hat{\theta}^{[1]} = (\hat{\theta}_{j1}^{[1]}, \hat{\theta}_{j2}^{[1]})$ is selected in first iteration, model intercept is updated to $f^{[0]}(\mathbf{x}) + \hat{\theta}_{j1}^{[1]}$.
 - During training, intercept is adjusted M times to yield $f^{[0]}(\mathbf{x}) + \sum_{m=1}^M \hat{\theta}_{j1}^{[m]}$



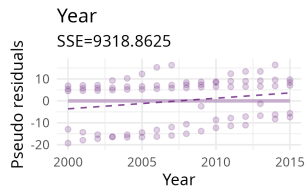
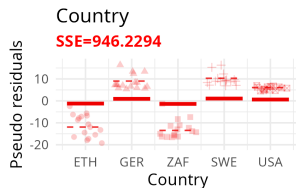
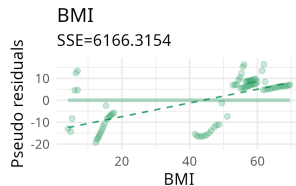
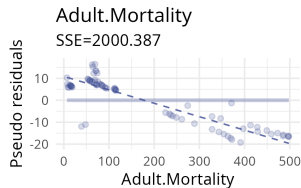
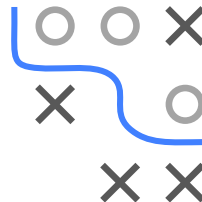
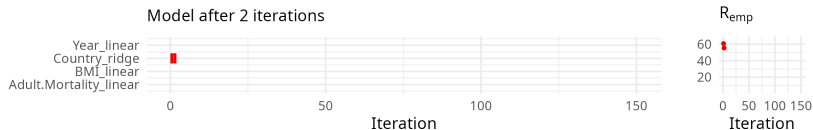
EXAMPLE: LIFE EXPECTANCY



— Partial feature effect - - - - Base learner fit to pseudo residuals

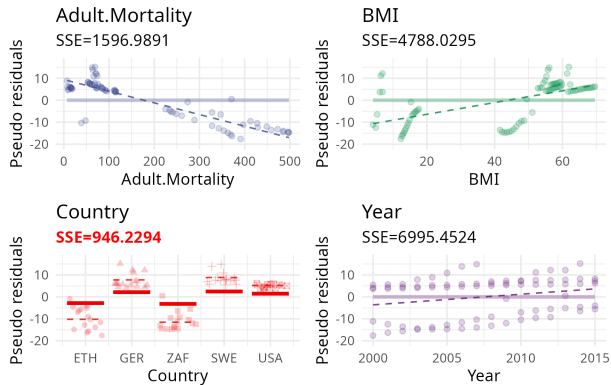


EXAMPLE: LIFE EXPECTANCY



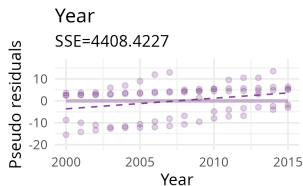
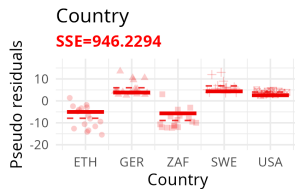
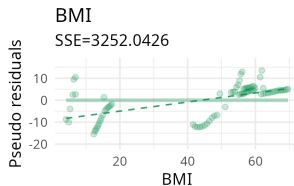
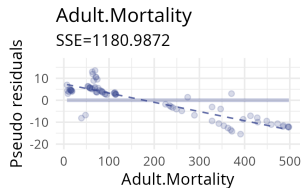
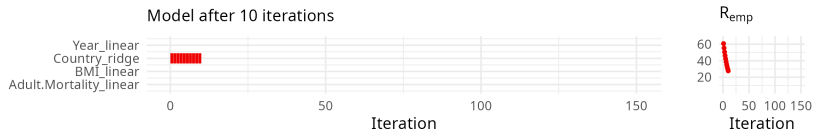
— Partial feature effect - - - - Base learner fit to pseudo residuals

A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path consists of the following cells: (0,0), (0,1), (1,1), (1,2), and (2,2). The cells (0,2), (1,0), and (2,0) are empty. The cells (1,0) and (2,0) contain a black 'X'. The cells (0,1) and (1,1) contain a grey circle. The cell (2,1) contains a grey circle.

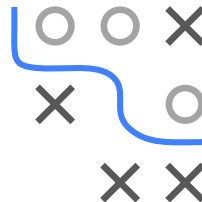


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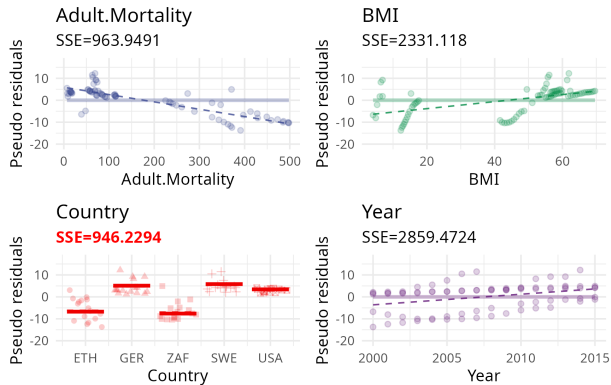
EXAMPLE: LIFE EXPECTANCY



— Partial feature effect - - - Base learner fit to pseudo residuals

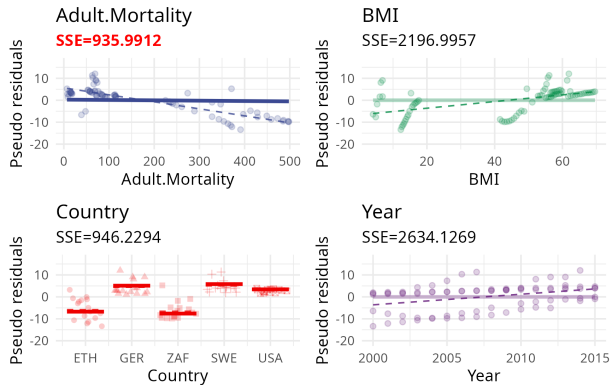


A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path is composed of three segments: a horizontal segment from (0,0) to (0,1), a vertical segment from (0,1) to (1,1), and a diagonal segment from (1,1) to (2,2). The cells (0,1), (1,0), (1,1), (1,2), and (2,1) are marked with a grey 'X', while the cells (0,2), (1,2), and (2,0) are marked with a grey circle.



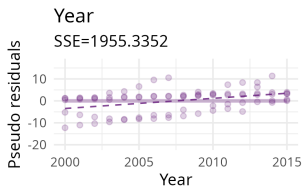
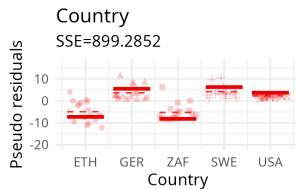
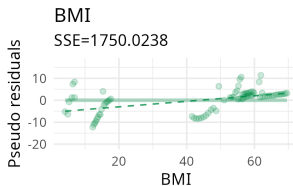
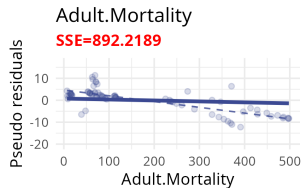
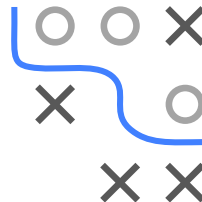
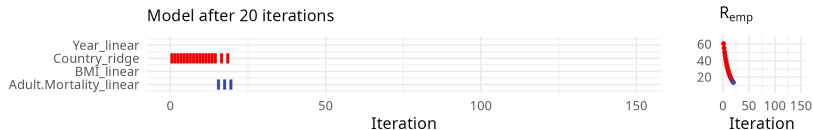
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A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path is composed of three segments: a horizontal segment from (0,0) to (0,1), a vertical segment from (0,1) to (1,1), and a diagonal segment from (1,1) to (2,2). The cells (0,1), (1,0), (1,1), and (2,2) are marked with a grey 'X', while the other cells are empty.



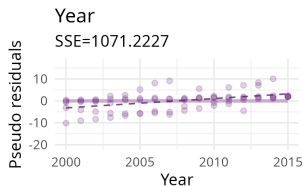
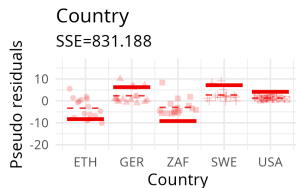
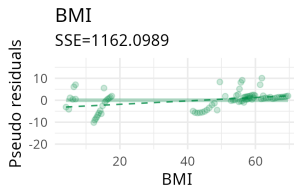
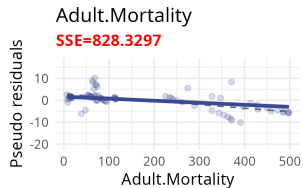
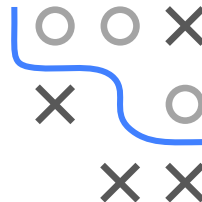
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EXAMPLE: LIFE EXPECTANCY

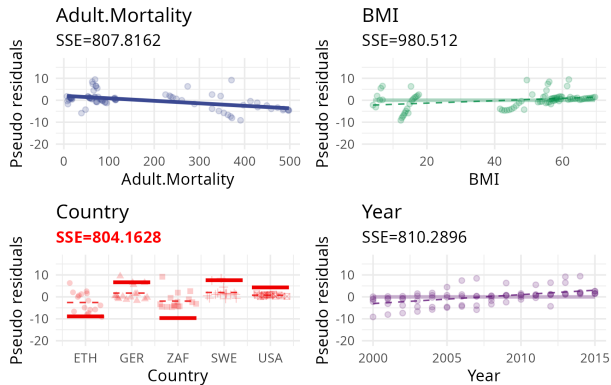


— Partial feature effect - - - - Base learner fit to pseudo residuals

EXAMPLE: LIFE EXPECTANCY

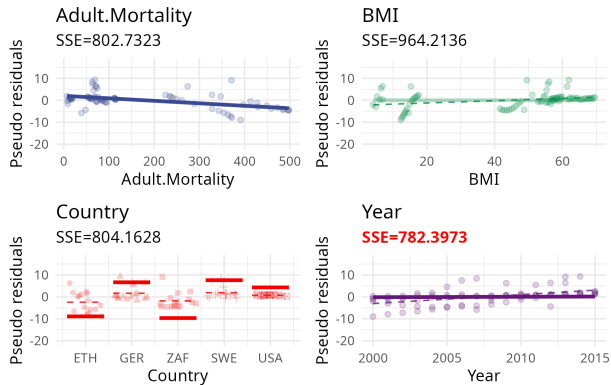


— Partial feature effect - - - - Base learner fit to pseudo residuals

[illegible]

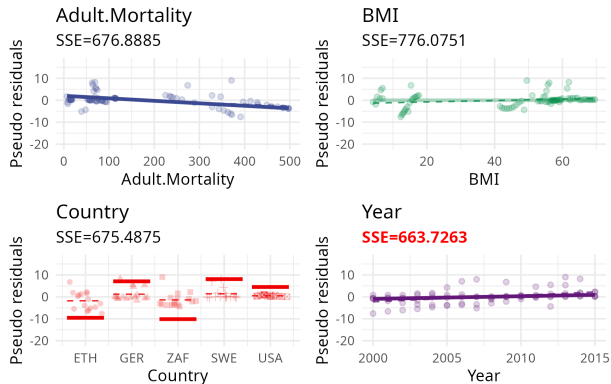
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A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path is composed of three segments: a horizontal segment from (0,0) to (0,1), a vertical segment from (0,1) to (1,1), and a horizontal segment from (1,1) to (2,1). The cells (0,2), (1,0), (1,2), and (2,2) are marked with 'X', while the other cells are empty.



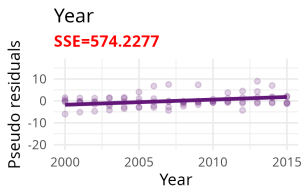
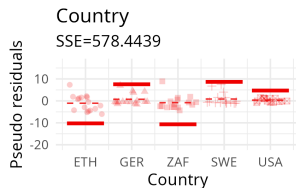
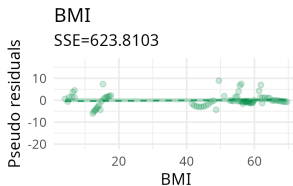
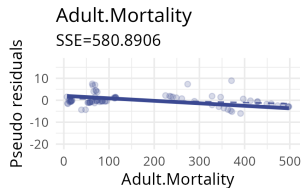
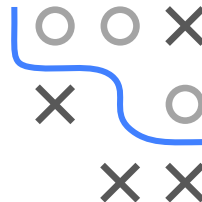
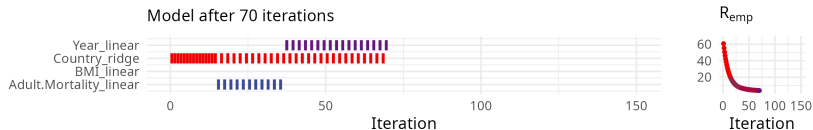
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A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path is composed of three segments: a horizontal segment from (0,0) to (0,1), a vertical segment from (0,1) to (1,1), and a diagonal segment from (1,1) to (2,2). The cells (0,1), (1,0), (1,1), (1,2), and (2,1) are marked with a grey 'X', while the cells (0,2), (1,2), and (2,0) are marked with a grey circle.



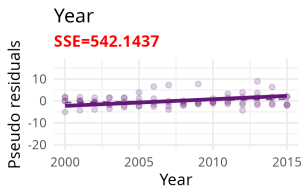
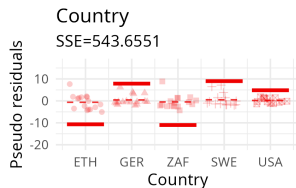
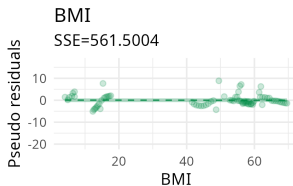
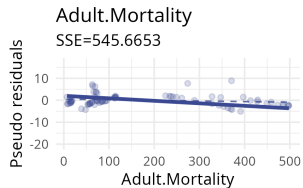
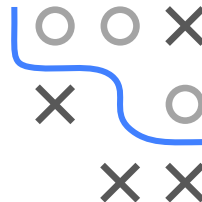
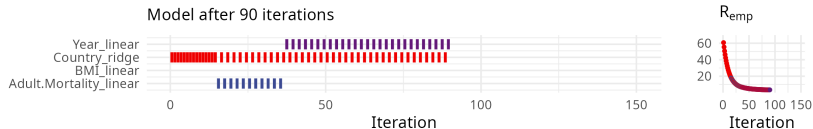
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EXAMPLE: LIFE EXPECTANCY



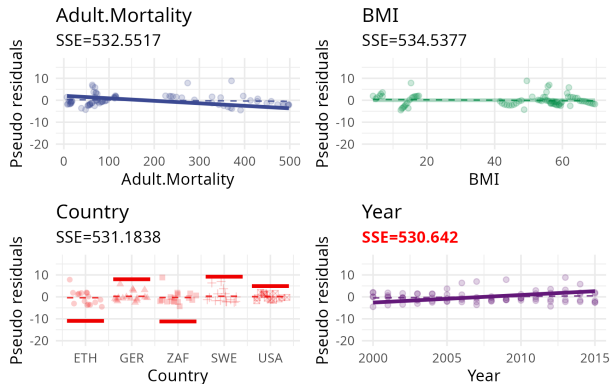
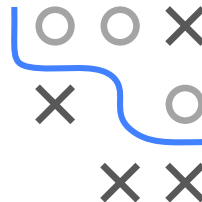
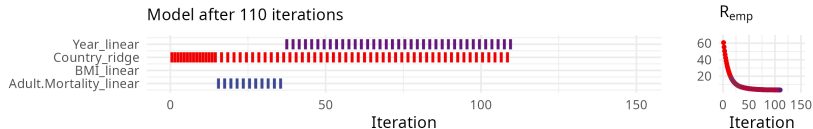
— Partial feature effect - - - - Base learner fit to pseudo residuals

EXAMPLE: LIFE EXPECTANCY



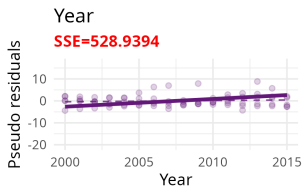
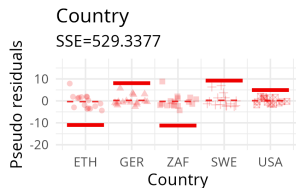
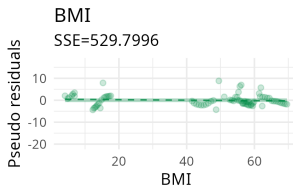
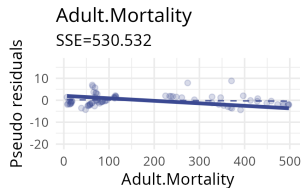
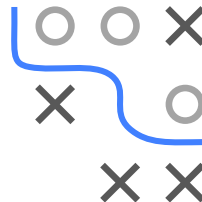
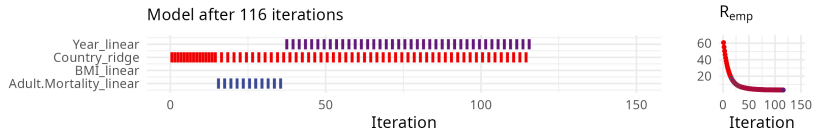
— Partial feature effect ···· Base learner fit to pseudo residuals

EXAMPLE: LIFE EXPECTANCY



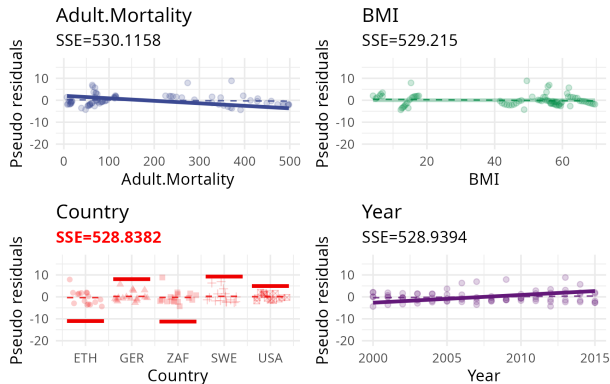
— Partial feature effect ···· Base learner fit to pseudo residuals

EXAMPLE: LIFE EXPECTANCY



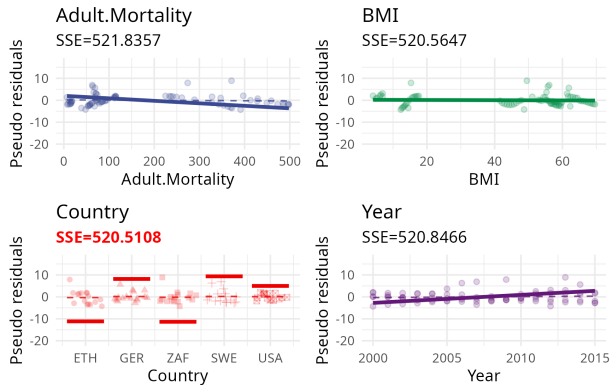
— Partial feature effect - - - - Base learner fit to pseudo residuals

A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path is composed of three segments: a horizontal segment from (0,0) to (0,1), a vertical segment from (0,1) to (1,1), and a diagonal segment from (1,1) to (2,2). The cells (0,1), (1,0), (1,1), and (2,0) are marked with a grey 'X', while the cells (0,2), (1,2), and (2,1) are marked with a grey circle.



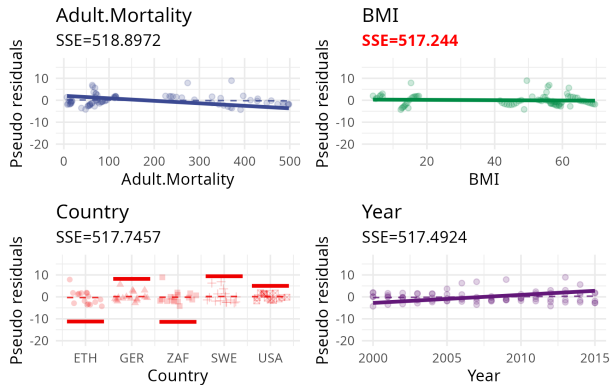
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A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path is composed of three segments: a horizontal segment from (0,0) to (0,1), a vertical segment from (0,1) to (1,1), and a diagonal segment from (1,1) to (2,2). The cells (0,1), (0,2), (1,0), (1,2), and (2,0) are marked with a grey 'X', while the cells (1,0) and (2,1) are marked with a grey circle.



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